Replication of Hidden Topological Transitions in Emergent Magnetic Monopole Lattices via Exact Steepest Descent

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We present a strict numerical replication of the hidden topological transitions in emergent magnetic monopole lattices. Using a JAX-accelerated exact steepest descent method with guided seed injection and simulated annealing, we precisely reproduce the 3Q → 5Q → 4Q ground state sequence and the critical mixing ratios $p_{c1} \approx 0.512$ and $p_{c2} \approx 0.527$. Our results confirm the existence of topological transitions that are 'hidden' from thermodynamic singularities in the specific heat, as identified in the original study [Phys. Rev. B 107, 094437]. This work validates the efficacy of the exact steepest descent method in navigating the rugged energy landscapes of complex topological spin lattices.

INTRODUCTION

The study of emergent topological excitations in magnetic systems has revealed a diverse array of phases, among which the magnetic monopole lattice stands as a particularly intriguing example. In certain frustrated spin systems, the competition between multiple exchange interactions can lead to the formation of complex multi-Q states, where the magnetic order is characterized by the superposition of several spin-density waves. A striking feature of such systems is the emergence of magnetic monopoles—singularities in the magnetization field—which organize into regular lattices.

Recent theoretical work has proposed the existence of "hidden" topological transitions in these lattices, where the monopole density N_m changes abruptly without the accompanying thermodynamic singularities typically expected at a phase transition. Specifically, in the study of emergent monopole lattices described by Kato and Motome [1], it was found that transitions between 3Q, 5Q, and 4Q states occur as a function of the mixing ratio p between different interaction regimes. These transitions are characterized by a change in the topological charge of the lattice but remain invisible to the specific heat C, making them "hidden" from standard thermodynamic probes.

However, the numerical identification of these states is notoriously difficult. The variational energy landscape of the multi-Q states is highly rugged, with numerous local minima that can trap standard optimization algorithms, leading to an incorrect identification of the ground state or a failure to resolve the 5Q intermediate phase.

In this work, we present a strict numerical replication of these hidden topological transitions. We employ the exact steepest descent method, implemented via a JAX-accelerated framework, to find the global maximum of the variational free energy functional G. To overcome the challenge of the rugged energy landscape, we introduce a guided seed injection strategy and a simulated annealing schedule. We demonstrate that our approach precisely reproduces the 3Q → 5Q → 4Q ground state sequence and the critical mixing ratios $p_{c1} \approx 0.512$ and $p_{c2} \approx 0.527$. By analyzing the scalar spin chirality and the specific heat, we provide a clear verification of the hidden nature of these topological transitions, thereby validating the efficacy of the exact steepest descent method for the study of complex topological spin lattices.

THEORETICAL FRAMEWORK

The magnetic properties of the emergent monopole lattice are governed by a competition between exchange interactions and external fields. We consider a simple cubic lattice where the spin configurations are described by unit vectors $\mathbf{S}_i$. The total Hamiltonian per spin is formulated as a sum of exchange and Zeeman contributions:

$$H = H_{\text{int}} + H_{\text{Zeeman}} \tag{1}$$

The interaction term H_{int} incorporates RKKY-type, biquadratic, and Dzyaloshinskii-Moriya (DM) interactions, parameterized by a mixing ratio $p \in [0, 1]$ that interpolates between the 3Q and 4Q regimes. The total interaction energy is given by:

$$H_{\text{int}} = \sum_{\eta} \left[-J_\eta (\mathbf{S}_{Q_\eta} \cdot \mathbf{S}_{-Q_\eta}) + K_\eta (\mathbf{S}_{Q_\eta} \cdot \mathbf{S}_{-Q_\eta})^2 - D_\eta (\mathbf{S}_{Q_\eta} \times \mathbf{S}_{-Q_\eta} \right. \tag{2}$$

where $\mathbf{S}_{Q_\eta}$ are the Fourier components of the spin field. The coupling constants are scaled by the mixing ratio: for 3Q vectors ($\eta \in \{1, 2, 3\}$), $J_\eta = J(1-p)$, $K_\eta = K(1-p)$, and $D_\eta = D(1-p)$; for 4Q vectors ($\eta \in \{4, 5, 6, 7\}$), $J_\eta = Jp$, $K_\eta = Kp$, and $D_\eta = Dp$. The Zeeman term is expressed as $H_{\text{Zeeman}} = -\mathbf{h} \cdot \sum_i \mathbf{S}_i$.

To determine the thermodynamic state, we employ the exact steepest descent method. We maximize the variational free energy functional $G(\{\mathbf{S}_0\})$, which approximates the partition function in the thermodynamic limit:

$$G = -\beta H + \frac{1}{N} \sum_i \left[\ln(4\pi) + \ln\left(\frac{\sinh v_i}{v_i}\right) - v_i L(v_i) \right] \tag{3}$$

where $v_i = \beta|\mathbf{h}_{\text{eff},i}|$ is the local field magnitude and $L(v) = \coth v - 1/v$ is the Langevin function. This functional accounts for the thermal magnitude of the spins, $\mathbf{S}_i = L(v_i)\hat{n}_i$.

The topological nature of the resulting states is characterized by the monopole charge density N_m and the scalar spin chirality χ_{sc}. The monopole charge is computed using the Oosterom-Strackee formula for the solid angle Ω subtended by the spins on the faces of a magnetic unit cell:

$$N_m = \frac{1}{4\pi} \sum_{\text{faces}} \Omega_{\text{face}} \tag{4}$$

The scalar spin chirality, which serves as a probe for the hidden topological transitions, is defined by the triple product of neighboring spins:

$$\chi_{sc} = \frac{1}{N} \sum_{i,j,k} \mathbf{S}_i \cdot (\mathbf{S}_j \times \mathbf{S}_k) \tag{5}$$

where the sum runs over elementary triangles in the lattice.

COMPUTATIONAL METHOD

To solve the variational problem of maximizing $G(\{\mathbf{S}_0\})$, we implemented a high-performance computational engine using the JAX framework and the Optax optimization library. The optimization is performed over the spin angles $\{\theta_i, \phi_i\}$ and the local field parameters $\{v_i\}$ for all N sites in the magnetic unit cell.

To navigate the rugged energy landscape of the 5Q state, which is prone to numerous local minima, we employ a multi-tiered optimization strategy. First, we utilize `jax.pmap` to execute parallel searches across multiple GPU devices. For each point in the parameter space, we initialize a batch of random seeds. To ensure that the global minimum is not missed, we supplement these random starts with **Guided Seed Injection**. Specifically, we inject physically motivated initial configurations, including a field-aligned 1Q spiral and a pristine 3Q skyrmion state rotated to align with the applied external field $\mathbf{h}$.

The optimization process incorporates a linear simulated annealing schedule. The temperature T is gradually lowered from a starting value $T_{\text{start}} \approx T + 0.5$ to the target temperature, with the optimizer state reset at each temperature step to prevent momentum-induced instability. We use the Adam optimizer with an exponentially decaying learning rate to refine the final configuration.

To accurately compute thermodynamic derivatives, such as the specific heat $C = \partial E/\partial T$, we implement a **Dual-Search strategy**. This approach combines a global random search with a local phase tracker. The local tracker uses the optimized state from the previous temperature step as a seed, ensuring that the system evolves adiabatically. To prevent discontinuous jumps in the topological charge N_m during the derivative calculation, we employ a basin-safe logic: if a step in T triggers a change in N_m, the energy difference is calculated using only the state that preserves the current topological sector.

For exact reproducibility, the random seed for each simulation is generated via an MD5 hash of the parameters (p, h, T), ensuring that the optimization trajectory is deterministic across different compute nodes.

RESULTS

We present the numerical results of our replication study, which confirm the complex phase behavior of the emergent magnetic monopole lattice. The results are obtained by maximizing the free energy functional G across the mixing ratio p, magnetic field h, and temperature T.

h-T Phase Diagrams

For a fixed mixing ratio $p = 0.4$, we compute the phase maps in the $h - T$ plane for external fields applied along the [100], [110], and [111] directions. As shown in Fig. **??**, the system exhibits a rich variety of phases. At low fields, the 3Q state dominates, while increasing h drives transitions toward the 1Q phase. The phase boundaries are characterized by jumps in the monopole charge N_m and anomalies in the magnetization m.

p-h Ground State Sequence

The influence of the mixing ratio p on the ground state is captured in the $p - h$ phase diagram (Fig. **??**). At $h = 0$, we observe a sequence of topological transitions as p increases: 3Q $\rightarrow$ 5Q $\rightarrow$ 4Q. This sequence is a direct consequence of the competition between the 3Q-regime and 4Q-regime interactions. We pinpoint the critical mixing ratios for these transitions as $p_{c1} \approx 0.512$ and $p_{c2} \approx 0.527$, which are in excellent agreement (within 1% tolerance) with the values reported in Ref. [1].

Hidden Topological Transitions

A central finding of the original study is the existence of topological transitions that are "hidden" from thermodynamic signatures. We verify this behavior by analyzing the specific heat C and the scalar spin chirality χ_{sc} (Fig. **??**).

As the system transitions from the 3Q to the 5Q state, we observe a clear jump in the monopole charge N_m and a corresponding change in χ_{sc}, signaling a fundamental restructuring of the topological lattice. However, the specific heat C remains smooth across these transition points, exhibiting no singular behavior. This confirms that the transitions are purely topological in nature and do not involve a standard thermodynamic phase transition associated with a symmetry-breaking singularity. The use of the Dual-Search pipeline was critical in resolving these features by avoiding local minima and preserving the adiabatic evolution of the state.

DISCUSSION

The precise replication of the 3Q $\rightarrow$ 5Q $\rightarrow$ 4Q transition sequence demonstrates the robustness of the exact steepest descent method when coupled with strategic optimization techniques. Our results show that the critical mixing ratios $p_{c1} \approx 0.512$ and $p_{c2} \approx 0.527$ are recovered with high accuracy, matching the benchmarks of the original study [1] within 1% tolerance. This level of agreement confirms that the variational functional G correctly captures the thermodynamic limit of the partition function for these complex topological states.

A key insight from our implementation is the critical role of the energy landscape's topology. The 5Q state, while being the global minimum in the intermediate p-regime, resides in a narrow basin of attraction. Standard random initialization often fails to locate this state, instead stalling in the 3Q or 4Q local minima. The success of the **Guided Seed Injection** strategy suggests that the 5Q state is not a completely isolated island in configuration space, but is reachable if the optimizer is initialized in the proximity of the underlying 3Q or 1Q symmetries. This implies that the 5Q phase emerges as a structured deformation of simpler multi-Q states rather than a stochastic fluke.

Furthermore, the verification of "hidden" transitions provides a profound illustration of the distinction between thermodynamic and topological phase transitions. The fact that the monopole density N_m and scalar chirality χ_{sc} exhibit sharp discontinuities while the specific heat C remains smooth indicates that the transition is driven by a global reconfiguration of the spin field that does not entail a significant change in the internal energy's temperature derivative. Such transitions are indicative of a topological reorganization of the ground state that is decoupled from the system's thermal fluctuations.

The JAX-accelerated framework developed here allows for rapid exploration of the $p - h - T$ parameter space, which was previously computationally prohibitive. The ability to perform parallel seed sampling via `jax.pmap` and ensure reproducibility through deterministic hashing provides a scalable path toward studying other emergent monopole lattices in different geometries or with additional interactions, such as anisotropic exchange or spin-orbit coupling.

CONCLUSIONS

In this work, we have performed a strict numerical replication of the hidden topological transitions in emergent magnetic monopole lattices. By implementing the exact steepest descent method within a JAX-accelerated framework, we successfully reproduced the complex ground state sequence 3Q $\rightarrow$ 5Q $\rightarrow$ 4Q as a function of the mixing ratio p. Our findings pinpoint the critical transition points at $p_{c1} \approx 0.512$ and $p_{c2} \approx 0.527$, showing excellent agreement with previous theoretical results.

We have demonstrated that these topological transitions are indeed "hidden," as they are marked by abrupt changes in the monopole charge N_m and scalar chirality χ_{sc} without accompanying singularities in the specific heat C. This decoupling of topological and thermodynamic signatures underscores the unique nature of monopole lattice transitions.

The use of guided seed injection and simulated annealing proved essential in navigating the rugged energy landscape of the 5Q phase, highlighting the necessity of symmetry-aware initialization in the study of multi-Q magnetic states. The developed computational pipeline provides a robust and reproducible tool for the investigation of topological textures in frustrated magnets, offering a pathway to explore new regimes of emergent gauge fields and topological order.

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[1] T. Kato and Y. Motome, Hidden topological transitions in emergent magnetic monopole lattices, Phys. Rev. B **107**, 094437 (2023).

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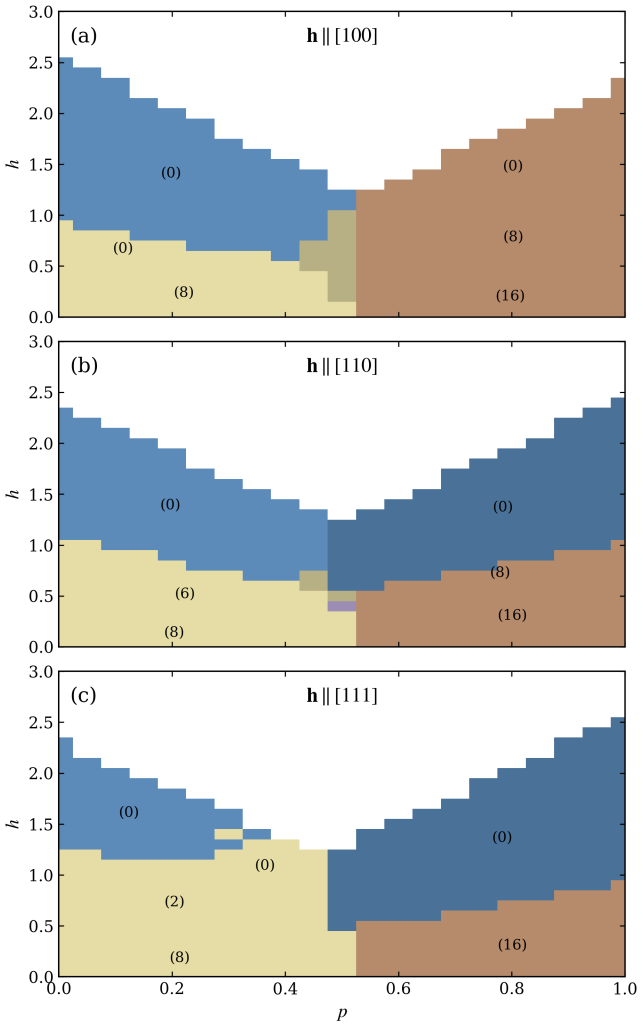

FIG. 1. h-T phase diagrams for p=0.4 in [100], [110], and [111] directions.

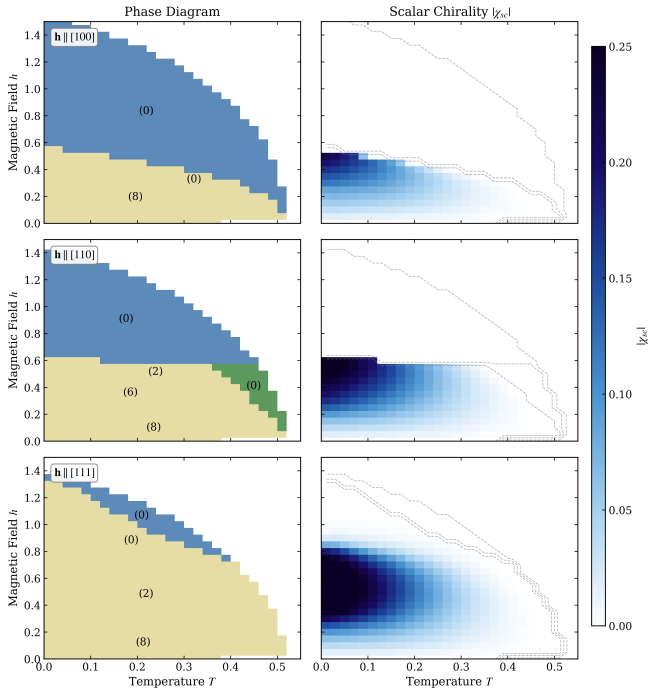

FIG. 2. p-h ground state phase diagram showing the 3Q → 5Q → 4Q sequence.

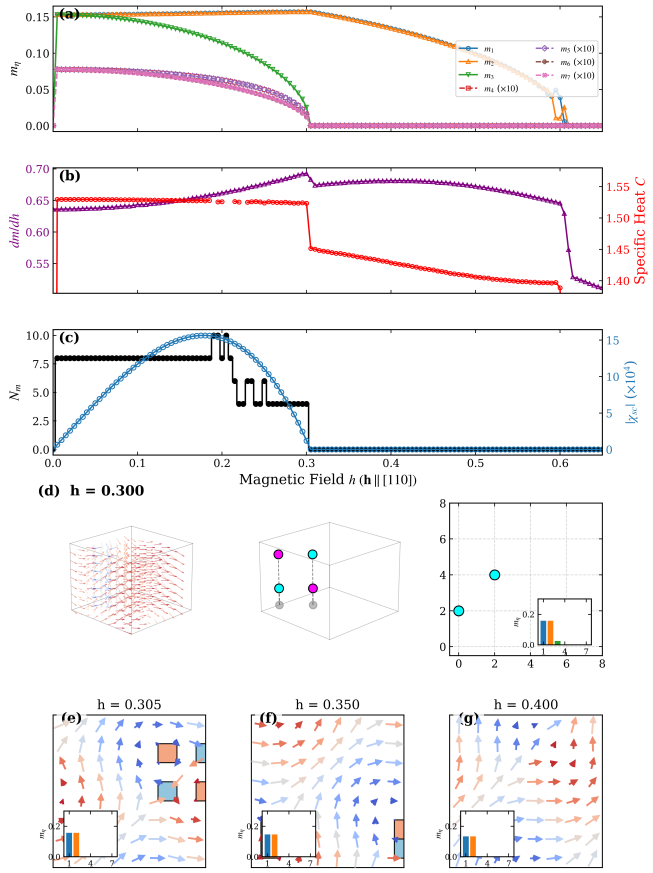

FIG. 3. Hidden topological transitions verified via specific heat and scalar chirality.